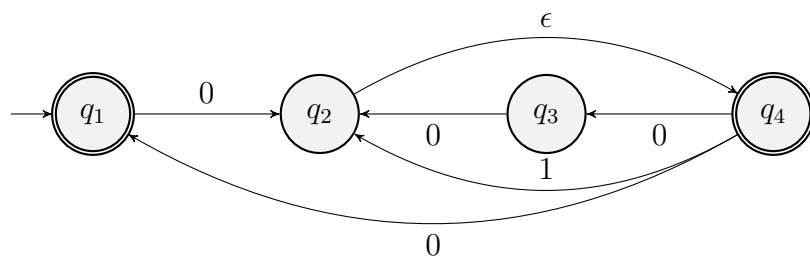


1 Question (10 points)



Use the NFA to DFA conversion algorithm to convert the given NFA above to a DFA.

2 Question (15 points)

Given the input alphabet $\{0, 1\}$. Give state diagram of DFA of $L = \{w \mid w \text{ does not contain the substring } 001\}$

3 Question (15 points)

Given the languages are binary, is the following language regular? Prove it. Let A be regular, then the language $A^R = \{w^R | w \in A\}$, where w^R is the string written w in reverse order.

4 Question (10 points)

Convert the following regular expression to NFA.

$$((aa \cup ba)^*ab)^*$$